PORYAAAA AUDIO ENGINE

What Was Wrong With the mGBA Audio Implementation

A source-backed diagnosis of the PSG, timing, mixing, resampling, and measurement defects that prevented poryaaaa from sounding like the ROM running through mGBA.

Bottom line: the mismatch was not one bad square-wave volume constant. The implementation diverged at several coupled boundaries - MP2K register behavior, hardware envelope timing, oscillator clocking, SOUNDBIAS cadence, output filtering, and the comparison harness itself.

12	19/19	8/8
confirmed implementation or measurement defects	audible isolated song/channel pairs with a strict local match	isolated stereo routing signatures

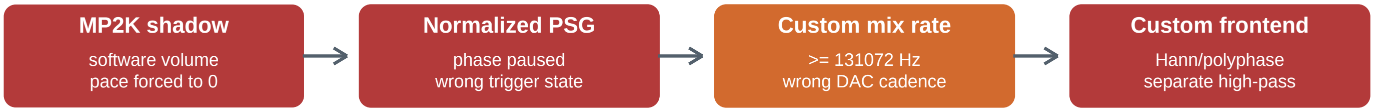
Scope

This report covers the uncommitted parity work in `packages/poryaaaa` as of 2026-07-10. It distinguishes corrected engine defects from the remaining comparison-indexing and combined PSG/program-wave validation boundary.

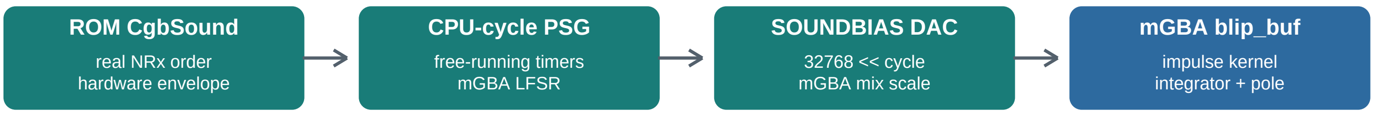
01 Executive diagnosis

The previous engine produced recognizable MP2K audio, but its PSG output did not pass through the same state transitions, clock domains, or frontend as mGBA. Several local approximations compounded into audible differences, especially in square-channel level and envelope behavior.

Previous implementation



Corrected implementation



The main failure pattern

- MP2K's software shadow envelope was treated as the final audio envelope instead of programming the Game Boy hardware envelope correctly.
- Square oscillators were represented by normalized phase increments rather than mGBA's integer GBA CPU-cycle deadlines.
- The complete chip mix ran at a convenient internal rate instead of the SOUNDBIAS-selected DAC cadence.
- A custom windowed-sinc/high-pass frontend replaced the exact blip_buf path heard from mGBA.
- The validation tool originally accepted polarity inversion and selected the strongest convenient window, which could overstate parity.

> **The fixes made the isolated waveform path converge. A controlled Weather voice 81 SQ2 pair reached 0.9999990187 correlation, 0.1401% residual, and +0.00114 dB. The three former song-level SQ2 outliers were comparison-indexing artifacts, not engine discrepancies.**

02 Confirmed engine defects

Area	What was wrong	Why it mattered	Correction
CGB gain	An artificial CGB gain multiplier was present.	It obscured the real PSG/mix scaling and encouraged tuning by ear.	Removed the multiplier and matched mGBA's integer PSG mix scale.
Track volume	MIDI tracks defaulted to 100 instead of MP2K's full 127.	Tracks without an early CC7 event started too quiet.	Set raw and effective defaults to 127 and retained song-volume scaling.
NRx2 envelope	Pace was forced to 0 while poryaaaa software-stepped volume.	mGBA's 64 Hz hardware envelope did not run like the ROM expects.	Emit attack/decay/release pace and direction; clock hardware envelopes at 64 Hz.
Register order	CGB start, pitch, pan, envelope, and trigger writes were in the wrong order.	Repeated writes reached different intermediate hardware states.	Port the observable CgbSound order, including pitch before volume for square/noise.
Length control	NRx4 length-enable was always cleared.	Voices with nonzero ToneData.length could not expire like the ROM.	Carry nonzero voice length into the hardware control write.
Trigger/off	Square/noise trigger and oscillator-off behavior was simplified.	Duty/LFSR/envelope state diverged across releases and retriggers.	Use the ROM's trigger-on-volume-write and CgbOscOff register sequences.

Why these defects reinforced one another

A wrong starting volume could appear to be a mixer problem; a wrong trigger could reset envelope state; a wrong register order could then make the same byte sequence sound different. Correcting only one constant did not address the state machine that produced the audio.

Primary implementation boundary

`packages/poryaaaa/plugin/m4a/m4a_cgb.c`

`packages/poryaaaa/plugin/m4a/m4a_internal.h`

03 Clocking and frontend defects

Area	What was wrong	Why it mattered	Correction
Square timer	A normalized 32-bit phase increment represented SQ1/SQ2 timing.	It did not preserve mGBA's exact elapsed CPU-cycle deadline behavior.	Advance duty indices every 16 * (2048 - frequency) GBA CPU cycles.
Silent phase	Square phase stopped while a channel was disabled.	Later register writes began from the wrong duty index.	Accumulate elapsed timer time across silence and catch up before writes.
Noise LFSR	Noise trigger/reset and shift behavior used the wrong sequence.	Noise timbre and repeated patterns differed from mGBA.	Use mGBA 0.10.5's width-specific LFSR state and shift/tap order.
NR52 phase	The post-m4aSoundInit frame sequencer began at step 0.	The first 512 Hz clock did not match mGBA's NR52 enable convention.	Initialize at frame step 7 so the first tick dispatches step 0.
DAC cadence	PSG/mix used a >= 131072 Hz internal-rate floor.	mGBA samples the complete GBA mix at the SOUNDBIAS cadence.	Render at 32768 << sampling_cycle: 32, 65, 131, or 262 kHz.
Frontend	A custom Hann/polyphase resampler and separate float high-pass were used.	The band-limit, ringing, clipping, latency, and DC decay differed.	Port mGBA's bundled blip_buf 1.1.0 impulse, integrator, clip, and 511/512 pole.

> **The decisive experiment was not another gain tweak: lowering the DAC cadence to 65536 Hz and using mGBA's blip impulse changed se_truck_stop SQ1 from roughly 0.966 correlation / 25.7% residual to approximately 0.99999 / 0.34% in the initial experiment.**

Primary implementation boundary

`packages/poryaaaa/plugin/hw_audio/hw_psg.c`

`packages/poryaaaa/plugin/hw_audio/hw_audio.c`

`packages/poryaaaa/plugin/hw_audio/hw_resample.c`

04 The measurement harness was also wrong

Early comparisons could produce a reassuring number without proving that both files contained the same waveform. That made it possible to compare levels across unrelated phases or envelope states.

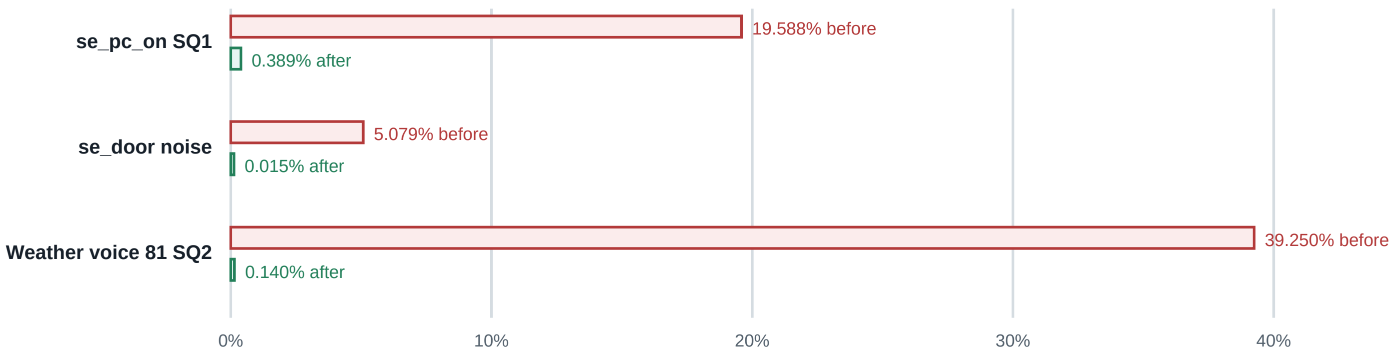
Area	What was wrong	Why it mattered	Correction
Polarity	Lag ranking and thresholds used absolute correlation.	A waveform multiplied by -1 could pass with near-zero fitted residual.	Rank and gate on signed positive correlation; inversion is a failure.
Window selection	The strongest passing window anywhere in the scan was selected.	A later convenient cycle could hide attack or envelope differences.	Select the earliest audible window that passes the strict shape gate.
Thresholds	Defaults were 0.98 correlation and 20% residual.	Square-like but materially different windows could authorize a dB claim.	Require >= 0.999 signed correlation and <= 5% residual.
Activity gate	Only the reference window had to be audibly active.	A nearly silent candidate could still produce a misleading ratio.	Require both windows above the PCM16 activity floor.
Scope	A local match risked being described as whole-song parity.	Repeated events at different envelope stages produced false song-level dB outliers.	Track a longer musical neighborhood before applying the strict local waveform gate.

Current proof sequence

- For timbre/level checks, fold both stereo files to mono using the arithmetic mean.
- Detect audible onset independently in each file.
- Track normalized 1.25-to-2-second energy neighborhoods with continuity constraints.
- Compare locally around the tracked neighborhood with a bounded sample-lag search.
- Remove DC only for shape analysis; do not normalize the captures.
- Fit gain and require positive correlation plus low residual before reading raw RMS dB.
- For routing checks, compare left and right independently and verify expected silence.

05 Evidence after correction

The residual chart shows how much of each candidate waveform remained unexplained after fitting one gain scalar. Lower is better. The Weather value is the controlled one-voice SQ2 pair, independent of song allocation and repeated-event indexing.



Representative before/after gain-fitted residuals. Values use different stages of the evolving harness and are diagnostic rather than a formal benchmark series.

775/777 direct C/C++ tests; two unrelated parity failures remain	**12/12** waveform and coverage tool tests	**PASS** headless mGBA smoke and paired-capture helper

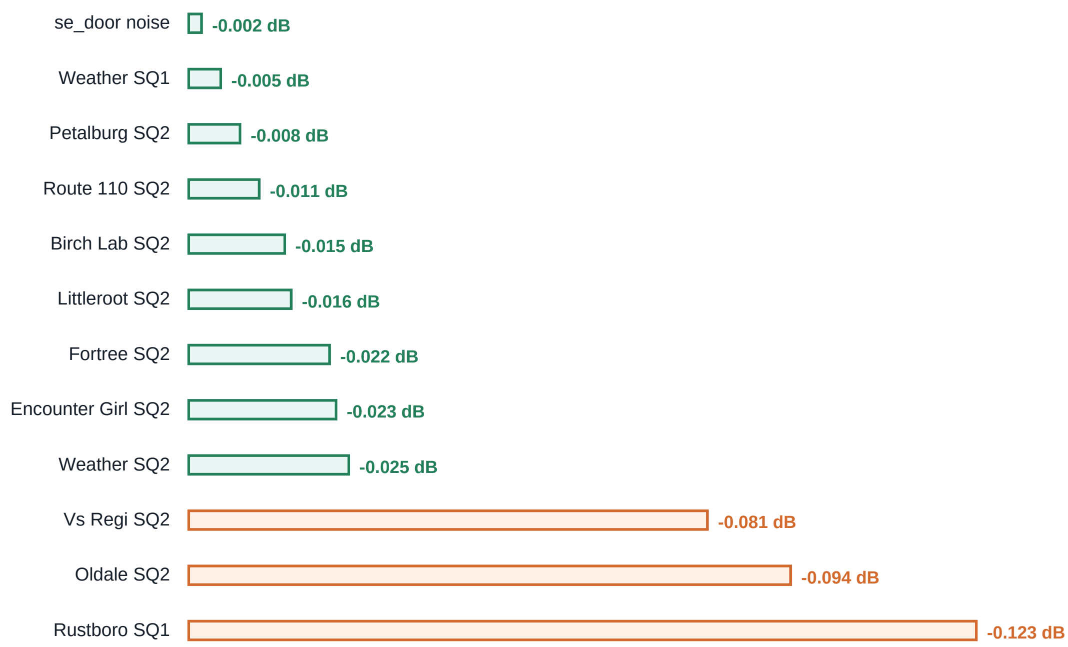
What the fixed baseline demonstrates

A ten-song matrix produced strict same-polarity local waveform matches for all 19 audible isolated SQ1/SQ2 pairs. Controlled Weather voice 81 and se_door noise also match at nearly identical raw RMS level. This rules out a universal PSG, SQ2, or noise gain defect.

> **The installed CLAP binary and the build artifact matched SHA-256 bf94ffb8dfb791a2c39a427d8c341081c8569f05f9a2f7462d91d044187d775d after the final build.**

06 Cross-song and stereo evidence

Ten representative mus_ songs were captured for eight seconds with SQ1 and SQ2 isolated. All 19 audible pairs found strict same-polarity local waveform matches. The chart shows neighborhood-aligned representative deltas; mus_littleroot SQ1 was silent and not comparable.



Representative raw RMS poryaaaa-to-mGBA deltas after neighborhood alignment and the strict local same-wave gate.

The former Encounter Girl, Weather Groudon, and Vs Regi SQ2 values of -0.368, -6.085, and -10.137 dB matched different repeated events or envelope stages. Neighborhood alignment gives -0.023, -0.025, and -0.081 dB.

Stereo routing also matches for isolated channels: Weather SQ1 is left-only, Weather SQ2 is right-only, Birch Lab SQ2 is left-only, and Door noise is centered. All 8 routing signatures passed; a deliberate channel swap failed and expected silent sides contained zero PCM.

07 Remaining validation boundary

What the evidence rules out

- A constant SQ2 gain error: controlled Weather voice 81 is within +0.00114 dB.
- Square duty tables or base frequency math: strict same-polarity waveform matches exist across duties 0 through 3.
- The global PSG mix and mGBA frontend: SQ1, SQ2, and noise controlled cases converge through the same path.
- Stereo collapse or reversal: isolated left-only, right-only, and centered captures reproduce mGBA routing.

What remains unproven

The current coverage reporter can choose different repeated musical events or envelope stages because it scans equal onset-relative offsets and accepts the earliest strict local shape match. The remaining engineering problem is robust musical-neighborhood indexing. The remaining audio-proof boundary is a combined PSG mix that includes the programmable-wave channel.

1	**Red fixture**	Repeat one square frequency at different envelope levels and expose the false earliest-window match.
2	**Neighborhood tracking**	Align normalized 1.25-to-2-second energy neighborhoods with continuity constraints.
3	**Strict local gate**	Require activity, positive correlation >= 0.999, and residual <= 5% near the tracked neighborhood.
4	**Combined PSG proof**	Compare left and right independently on a mix that includes the programmable-wave channel.

Next action: add a red repeated-event indexing fixture, implement the smallest neighborhood-aligned comparison mode, then use it to prove combined PSG/program-wave stereo. Do not reopen SQ2 allocation tracing without new correctly aligned evidence.

08 Code map and verification

Primary corrected modules

MP2K CGB register/envelope behavior	`packages/poryaaaa/plugin/m4a/m4a_cgb.c`
CGB channel state	`packages/poryaaaa/plugin/m4a/m4a_internal.h`
PSG oscillator/envelope/frame sequencer	`packages/poryaaaa/plugin/hw_audio/hw_psg.c`
SOUNDBIAS cadence and frontend orchestration	`packages/poryaaaa/plugin/hw_audio/hw_audio.c`
mGBA blip_buf port	`packages/poryaaaa/plugin/hw_audio/hw_resample.c`
Unit regressions	`packages/poryaaaa/test/test_engine.c`
Headless reference tools	`packages/poryaaaa/tools/mgba-reference/`
PSG parity handoff	`.scratch/mgba-sq2-parity/CONTEXT.md`

Reference sources used for parity decisions

- Pokemon Emerald/hearth-test compiled m4a implementation, especially src/m4a.c CgbSound and CgbOscOff.
- mGBA 0.10.5 src/gb/audio.c and src/gba/audio.c for PSG clocks, envelopes, sampling, mixing, and NR52 behavior.
- mGBA 0.10.5 bundled blip_buf 1.1.0 for the output impulse/integrator frontend.

Latest focused checks

Check	Result
Direct engine suite	775/777; two unrelated voicegroup-core parity failures
Comparator tests	6/6
Coverage tests	6/6
Paired-capture helper	Pass
Headless mGBA smoke	Pass
Isolated song matrix	19/19 audible SQ1/SQ2 pairs with strict local matches
Stereo routing	8/8 isolated routing signatures
git diff --check	Pass

Worktree note: the parity changes are uncommitted and coexist with unrelated modifications in poryaaaa-m4l, voicegroup-core, voicegroup-lsp, and other files. Review and stage surgically.